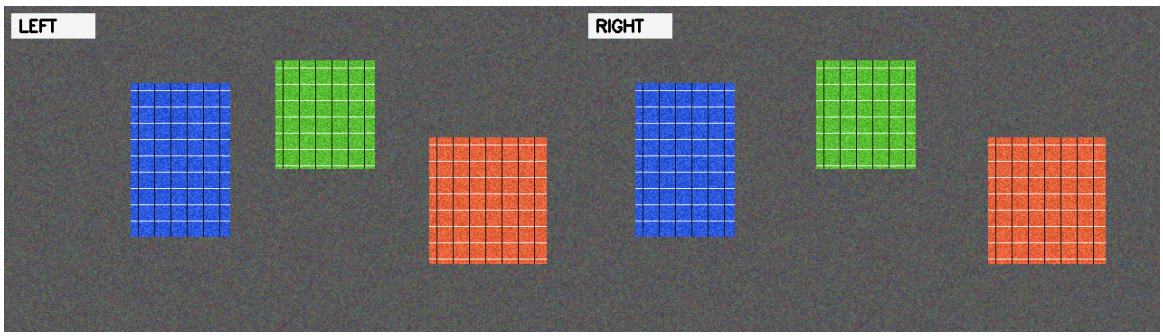
Sardar Vallabhbhai National Institute of Technology, Surat
Department of Computer Science and Engineering
M.Tech(CSE/DS) - I Sem - 2026-27
Computer Vision and Image Processing (CSCS111/CSDS119)
Lab Assignment 9 - Depth Analysis

Input: `stereo_left.png`, `stereo_right.png`
Problem: Estimate which objects are near or far using a rectified stereo pair. The images are already rectified, so corresponding points should lie on approximately the same horizontal rows.
Camera parameters: Focal length $f = 800$ pixels; baseline $B = 0.20$ metres. Approximate designed disparities are 80 pixels (near), 40 pixels (middle), and 20 pixels (far). Matching results may vary because the images are synthetic.



Left and right rectified images

1. Load and display the stereo pair in grayscale. Identify common objects and observe their horizontal displacement.

2. Compute a disparity map using StereoBM or StereoSGBM. Display the map and change one matching parameter to observe its effect.

3. Estimate depth using
$$Z = \frac{fB}{d},$$
where Z is depth in metres, f is focal length in pixels, B is baseline in metres, and d is disparity in pixels. Estimate the depth of at least two objects using representative disparity values.

challenge: Identify regions with missing or unreliable disparity. Explain why matching can fail around boundaries, textureless areas, or repeated patterns.
- Generate Stereo pair, disparity map, depth estimates for selected objects, and brief observations.
- Try cv2.StereoSGBM_create() or cv2.StereoBM_create(). Parameters are also listed in stereo_calibration.txt.